

# Seaborn Assignment

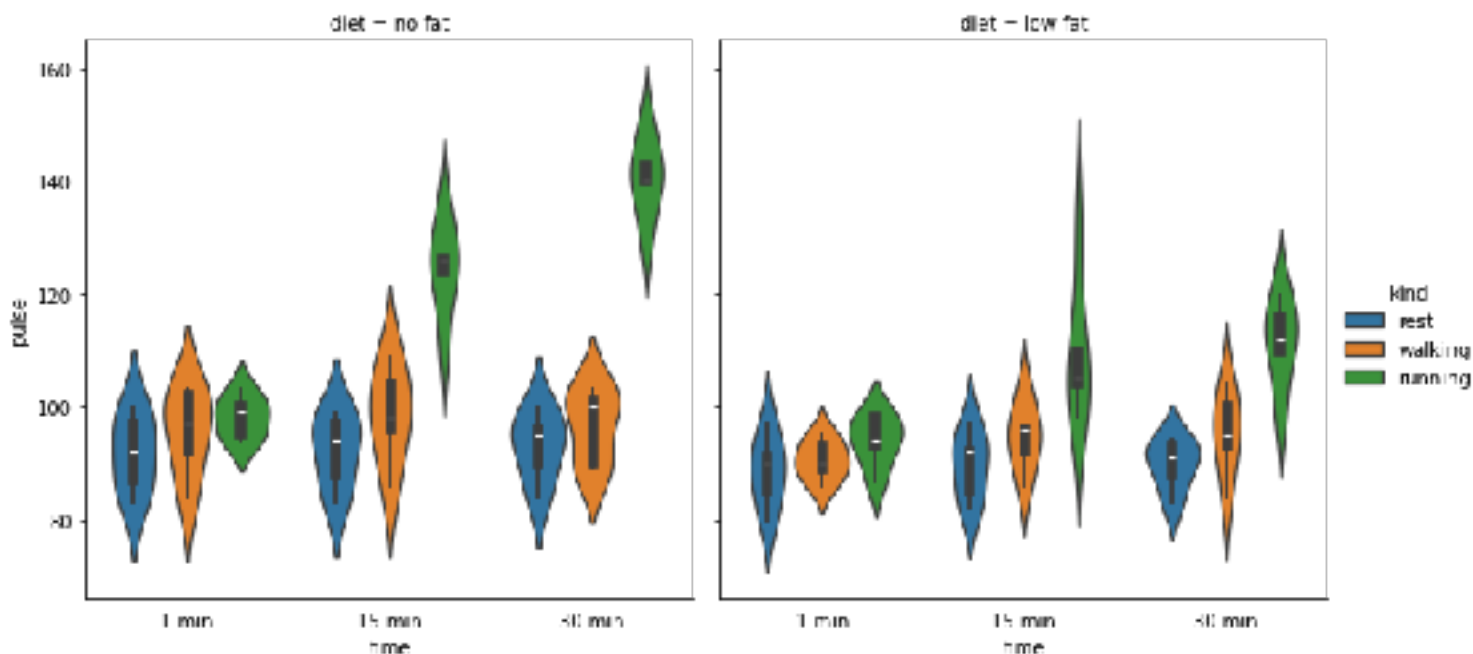
## Why do we go for swarmplot?

```
import seaborn as sns
sns.swarmplot()
```

In Swarmplot, the datapoints adjust along categorical axis so that the points do not overlap which ensures every single observation is visible, easier to see the true density of data. This helps in identifying Outliers or clusters that are hidden in summary. Also swarmplot is more effective for smaller datasets.

## Factor plot(cat plot)

```
df = sb.load_dataset('exercise')
sb.catplot(x = "time", y = "pulse", hue = "kind", kind = 'violin', col =
"diet", data = df);
```



### **Observations from the above factor plot(cat plot):**

- Over all people with no fat diet, the pulse rate is between 70-160 where only few are with pulse rate 160
- With low fat diet, the pulse rate is between 65-150 where few are with pulse rate between 120-150

### **In either of the diet:**

- People have higher pulse rate range between 80-160 while running than resting or walking.
- While resting people have pulse rate between 70-110 which is lowest compared to other activities.
- While walking people have pulse rate between 70-120 which is almost same as resting.

### **In diet with no fat,**

- In 1st min to 30th min, while resting there is no change in pulse rate which is 70-110
- While walking, from 1st min to 15th min the pulse rate maximum range increased from 115 to 120. Interestingly in 30th min, there is a drop in maximum level to 110 and increase in minimum level from 70 to 80.
- While running, from 1st min to 15th min the pulse rate minimum level increased from 90 to 100 and maximum range increased from 110 to 150. In 30th min, there is a increase in minimum level to 120 and maximum level to 160 which is the maximum of all.

### **In diet with low fat,**

- While resting, from 1st min to 15th min the pulse rate minimum level increased from 65 to 70 where there is no change in maximum level which is 115. In 30th min, there is a drop in maximum level to 100 and increase in minimum level to 75.

- While walking, from 1st min to 15th min, there is a drop in minimum level from 85 to 75 where maximum level increased from 100 to 110. In 30th min, the trend continues, with a drop in minimum level to 70 and maximum level increased to 115.
- While running, from 1st min to 15th min, there is a drop in minimum level from 80 to 75 where maximum level increased from 105 to 150. In 30th min, there is a drop in maximum level to 130 and increase in minimum level to 90.

Largely, majority of people's pulse rate lies between 90 to 110.